



Prob.1: Consider a series RLC circuit has $R = 10 \Omega$, $L = 20 \text{ mH}$, $C = 5 \text{ nF}$, and $V_s = 5/\underline{0^\circ} \text{ V}_{rms}$.

1. The resonant frequency (ω_0) in (rad/sec) is:
 - a. 35 K
 - b. 100 K
 - c. 20 K
 - d. None of all
2. The quality factor (Q) is:
 - a. 200
 - b. 50
 - c. 100
 - d. None of all
3. The cutoff frequencies ω_1 and ω_2 (in rad/sec) are:
 - a. (19.2, 20.8) K
 - b. (24, 26) K
 - c. (99.5, 100.5) K
 - d. None of all
4. The voltage across the capacitor (V_C) at resonance (in V) is:
 - a. 1000
 - b. 100
 - c. 500
 - d. None of all

Prob.2: Two Loads connected in parallel across a voltage source of $100/\underline{0^\circ} \text{ V}_{rms}$, where the loads are:

Load 1: absorbs 250VA at 0.6 lagging p.f., and Load 2: absorbs 200W at 0.8 leading p.f

5. The current of load 1 is
 - a. $1.3 - j0.75 \text{ A}$
 - b. $0.75 - j1.0 \text{ A}$
 - c. $1.5 - j2 \text{ A}$
 - d. None of all
6. The current of load 2 is
 - a. $1.3 - j0.75 \text{ A}$
 - b. $0.75 - j1.0 \text{ A}$
 - c. $2 + j1.5 \text{ A}$
 - d. None of all
7. The complex power supplied by the source is
 - a. $564.8 + j501.2 \text{ VA}$
 - b. $350 + j150 \text{ VA}$
 - c. $231.5 + j564.8 \text{ VA}$
 - d. None of all
8. The power factor of the equivalent load is
 - a. 0.99 lagging
 - b. 0.748 lagging
 - c. 0.38 leading
 - d. None of all
9. The current supplied by the voltage source in rms is
 - a. $2.822 - j2.506 \text{ A}$
 - b. 2.822 A
 - c. 3.535 A
 - d. None of all
10. The impedance of the equivalent load is
 - a. $24.87 - j60.67 \Omega$
 - b. $28 + j4 \Omega$
 - c. $60.67 + j24.87 \Omega$
 - d. None of all

Prob.3 : Consider the circuit shown in Figure 1 where $V = 10/\underline{-60^\circ} \text{ V}_{rms}$, and $I_s = 5/\underline{10^\circ} \text{ A}_{rms}$.
answer questions 11-13.

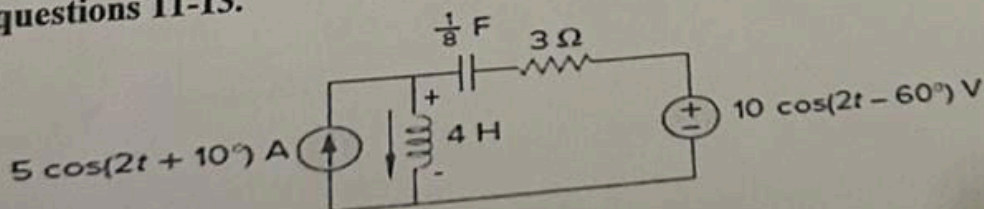


Fig.1

1. The voltage across the inductor due to V source only (in V) is:
 - a. $12.9 - j8.29$
 - b. $14.71 - j6.29$
 - c. $12 + j5$
 - d. None of all
2. The voltage across the inductor due to I_s source only (in V) is:
 - a. $-12.79 - j6.26$
 - b. 40
 - c. $39.76 - j4.76$
 - d. None of all
3. The current through the inductor due to both sources in A is:
 - a. $8.71 + j6.29$
 - b. $1.33 - j6.81$
 - c. $15.4 - j31$
 - d. None of all



Prob.4: A balanced abc-sequence Y-connected source with $V_{an}=120\angle 30^\circ$ V is connected to a balanced load $(24 + j19) \Omega$ per phase. The line joining the source and the load has an impedance of $(1+j) \Omega$ as shown in figure 2

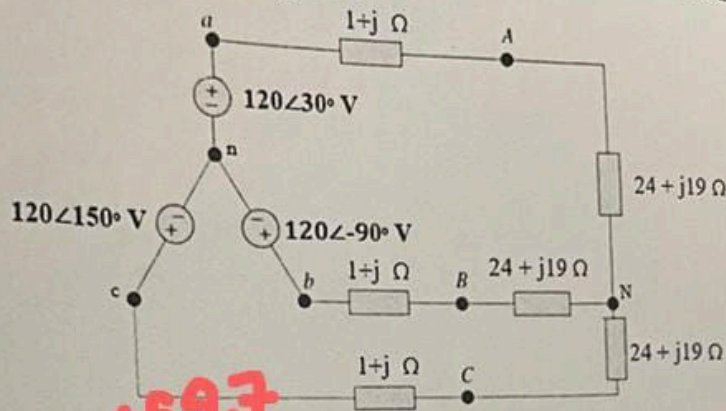


Fig. 2

1. The voltage V_{AB} is:

2. The line current I_a is:

3. The phase current I_b is:

4. The total average power supplied by the source is:

5. The total complex power consumed by the load is:

Prob.5: Consider the circuit shown in figure 3:

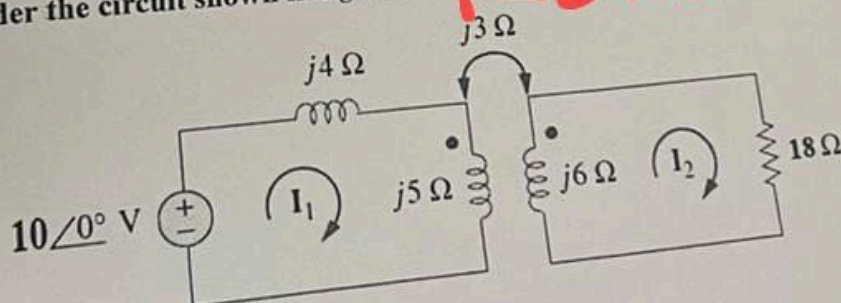


Fig. 3

1. The equivalent input impedance (Z_{in}) seen by the source (in Ω) is:

2. The current supplied by the source (I_1) is:

3. The power absorbed by the 12 Ω resistor is:

4. The required equations (without simplification) to solve the problem are: